SIGN-LANGUAGE

Spring 2026

Meeting No.4

System Development

& Datasets

What was done this semester

New models and datasets:

- Obtained the accuracy results of pretrained ViViT and SlowFast models with no fine-tuning on WLASL-100 and WLASL-300 datasets
- Completed full fine-tuning for ViViT and SlowFast models for WLASL-100 and WLASL-300 datasets
- Completed parameter efficient fine-tuning (PEFT) for SlowFast model for WLASL-100 and WLASL-300 datasets

System development:

- Developed ASL image dataset recognizing system using ResNet18 model
- The user interface, built using Tkinter library, allows for real-time display of the webcam feed, predicted signs, and user interaction features
- Trained ViViT module on WLASL videos and the system was created on first 10 classes

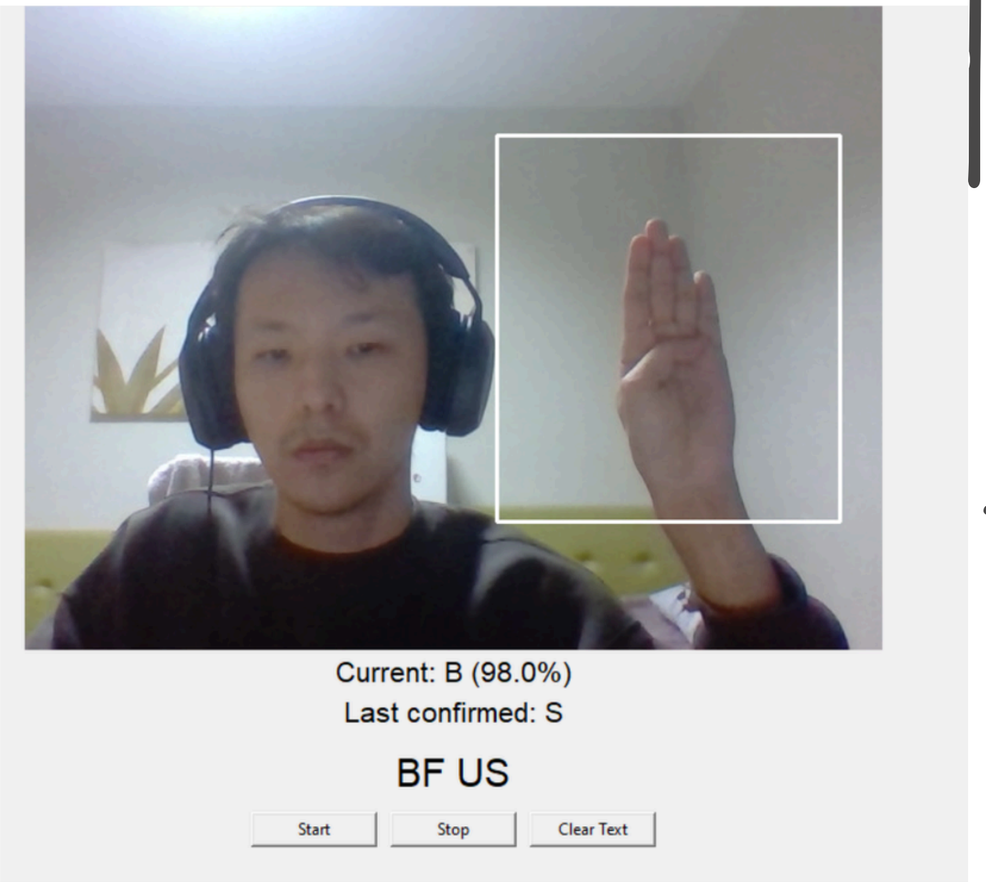
IMAGE DATASET IMPLEMENTATION

Dataset:

- Classes: 29 ASL signs (American Sign Language)
- Images: 30 images per class (every 100th image sampled for training)
- Source: ASL Alphabet dataset for training and validation

Model:

- Module: ResNet18
- Training: Fine-tuned on the ASL dataset using standard transfer learning practices
- Accuracy: Achieved high accuracy on both training and test datasets, with no overfitting observed



VIDEO DATASET IMPLEMENTATION

Dataset:

- Classes: 2000 words (ASL signs for words, instead of just letters)
- Videos: 5 to 10 videos per class, each lasting 1 to 4 seconds
- Source: Custom ASL word dataset for training and validation

Model:

- Module: ViViT
- Training: Fine-tuned on the ASL word dataset using video-based transfer learning
- Accuracy: Achieved high accuracy on training, but validation score got 30-35%. The reason is low amount of videos per class.

OLD IMPLEMENTATION

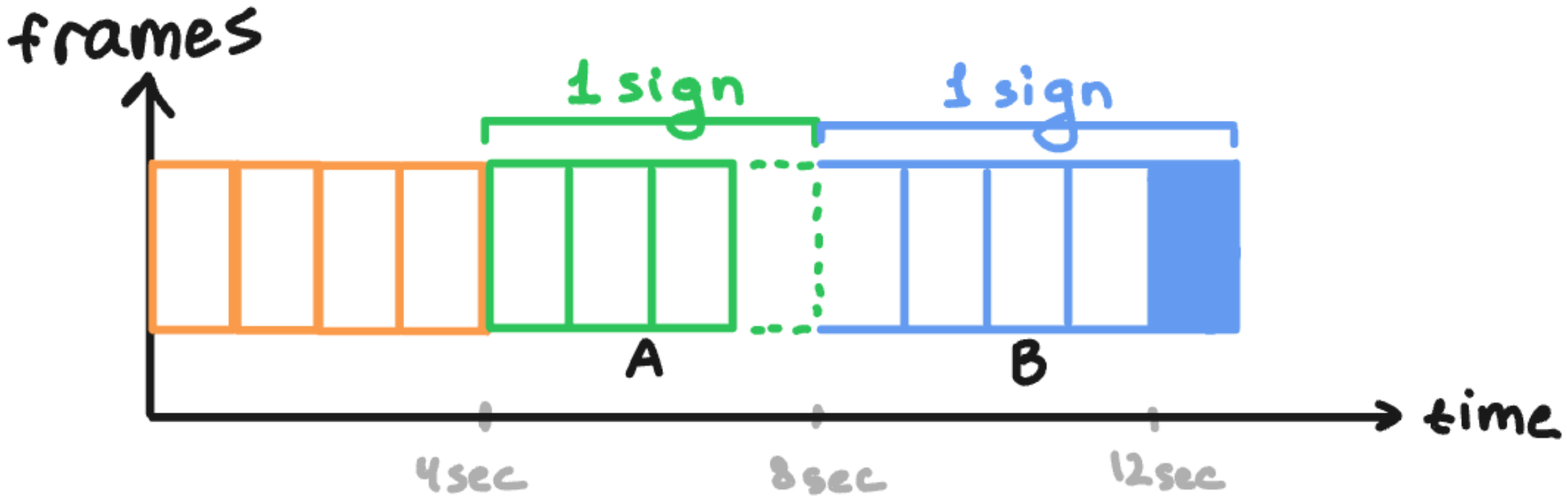
Long video example:
camera captured = 80 frames
model needs = 32 frames

Then:
80 -> evenly sample -> 32 frames

Short video example:
camera captured = 8 frames
model needs = 32 frames

Then:
np.arange(32) % 8 (video is looped)

SPECIFIC CASE HANDLING



Scenario A: Some signs may last for less than the 4-second threshold used for prediction. The system captures a buffer of frames (32) over time, allowing for partial signs to still be included in the prediction. The model will check the buffered frames after each 4-second interval to account for these shorter signs, ensuring they are still recognized even if they were held for less than the full threshold

Scenario B: Signs that last longer than 4 seconds may cause repeated predictions or redundant recognition. The system uses a release state that prevents multiple detections of the same sign. Once a sign is confirmed, the system waits until the sign is "released" before it can be detected again. This ensures that long-held signs are only recognized once, preventing overfitting to prolonged gestures.

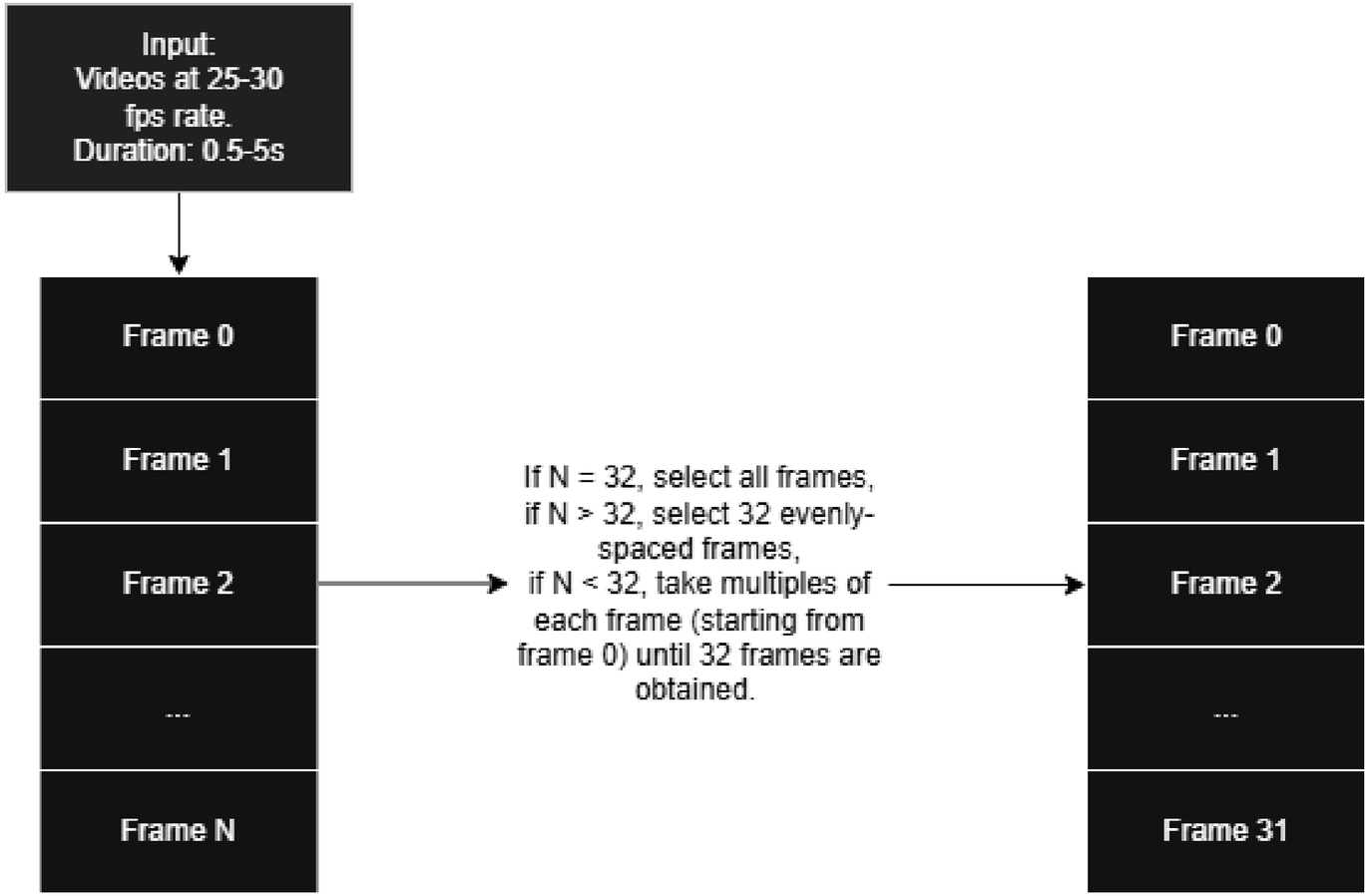
NEW IMPLEMENTATION

number of webcam frames = number of frames we took from the video = 32 frames

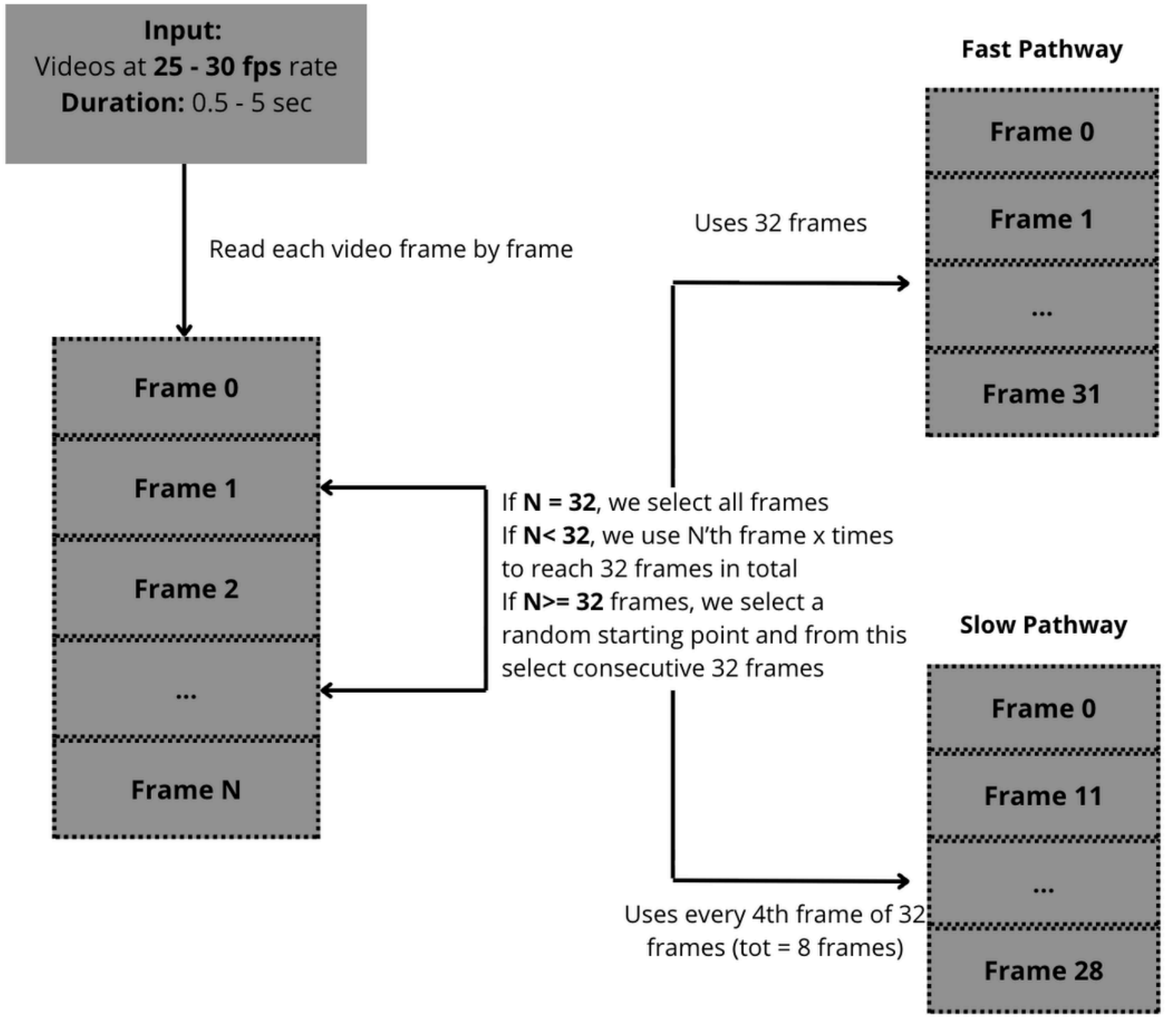
- Once the buffer reaches 32 frames, the frames form a single video clip.
- The clip length matches the ViViT model requirement (num_frames = 32).

- The system computes softmax probabilities.
- Displays the Top-3 predicted signs with confidence scores in real time.

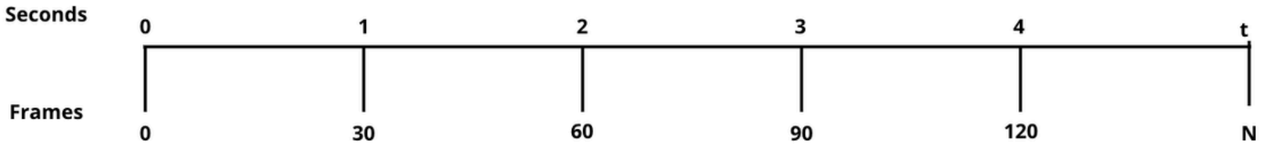
VIVIT FRAME HANDLING FOR BOTH TRAINING AND INFERENCE



SLOWFAST FOR TRAINING ON WLASL



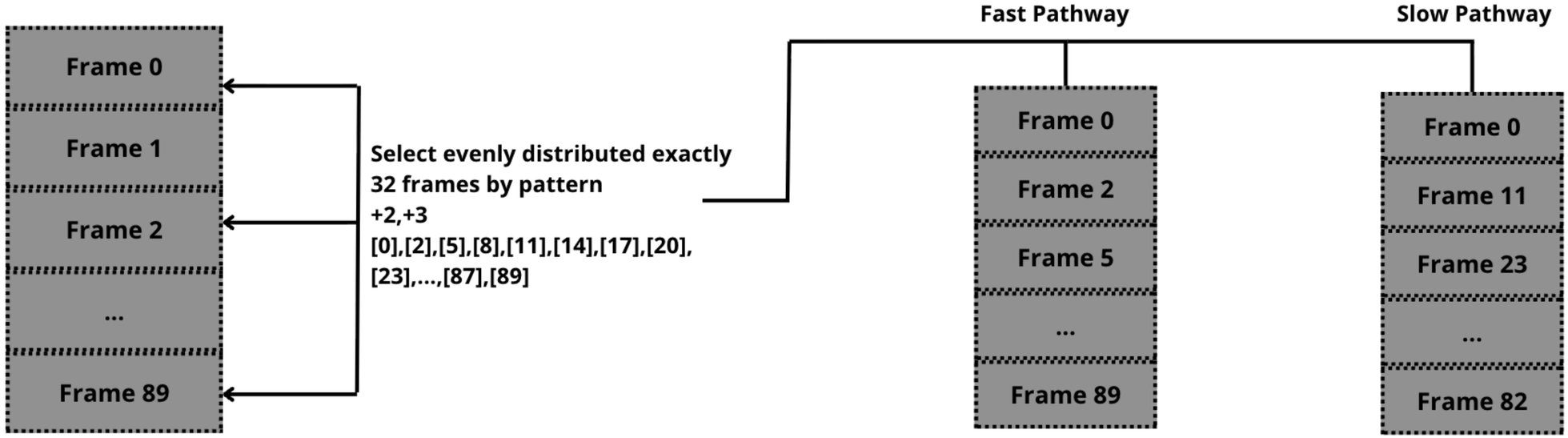
IMPLEMENTING SLOWFAST FOR SYSTEM



Sliding Window

Time →
Window 1 (0-3s): [Frame 1 → Frame 90]
Window 2 (0.1-3.1s): [Frame 4 → Frame 93]
Window 3 (0.2-3.2s): [Frame 7 → Frame 96]
Overlap: 87 frames common between windows
Processing: Every 100ms (new 3 frames) → new prediction
Keep last 3 seconds in buffer

Unifrom Sampling for frames selection



Available accuracy results

Model + fine tuning technique	WLASL_100	WLASL_300
SlowFast (Full Fine Tuning)	54.26%	38.27%
SlowFast(Lora Fine tuning)	46.12%	42.65%
ViViT (Full Fine Tuning)	56.2%	32.9%
SlowFast (No Fine Tuning)	7.8%	1.5%
ViViT (No Fine Tuning)	27.5%	0.2%

GitHub Link:

AbdollaZ/
seniorproject_signlangu...

This is a repository for hosting CSCI 409 Senior Project documentation and files. Project topic is Sign Language Capturing via...

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AbdollaZ/seniorproject_signlanguage: This is a repository for hosting CSCI 409 Senior Project documentation and files. Project topic is...

DATASET STATISTICS

WLASL (100 CLASSES)

	fps	frames	duration_sec
count	1443.000000	1443.000000	1443.000000
mean	28.490785	70.146223	2.465573
std	2.552664	26.078514	0.924995
min	20.000000	19.000000	0.767433
25%	25.000000	50.000000	1.793456
50%	29.970030	70.000000	2.435767
75%	29.970030	87.000000	2.969603
max	50.000000	195.000000	8.125000

AVERAGE FPS: 28.49078503048832
MIN FPS: 20.0
MAX FPS: 50.0

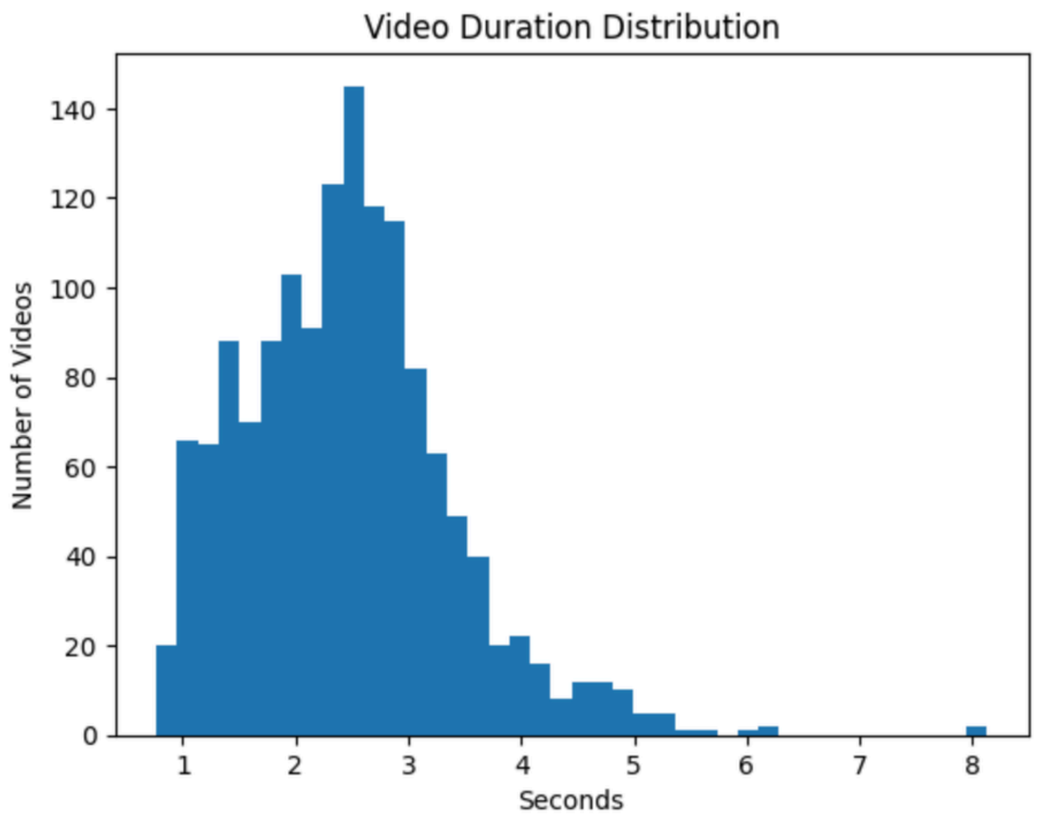
WLASL (300 CLASSES)

	fps	frames	duration_sec
count	3667.000000	3667.000000	3667.000000
mean	28.412865	70.623671	2.490486
std	2.695074	26.345115	0.935852
min	12.000000	19.000000	0.700701
25%	25.000000	50.000000	1.801802
50%	29.970000	71.000000	2.466333
75%	29.970030	87.000000	2.969633
max	50.000000	232.000000	8.125000

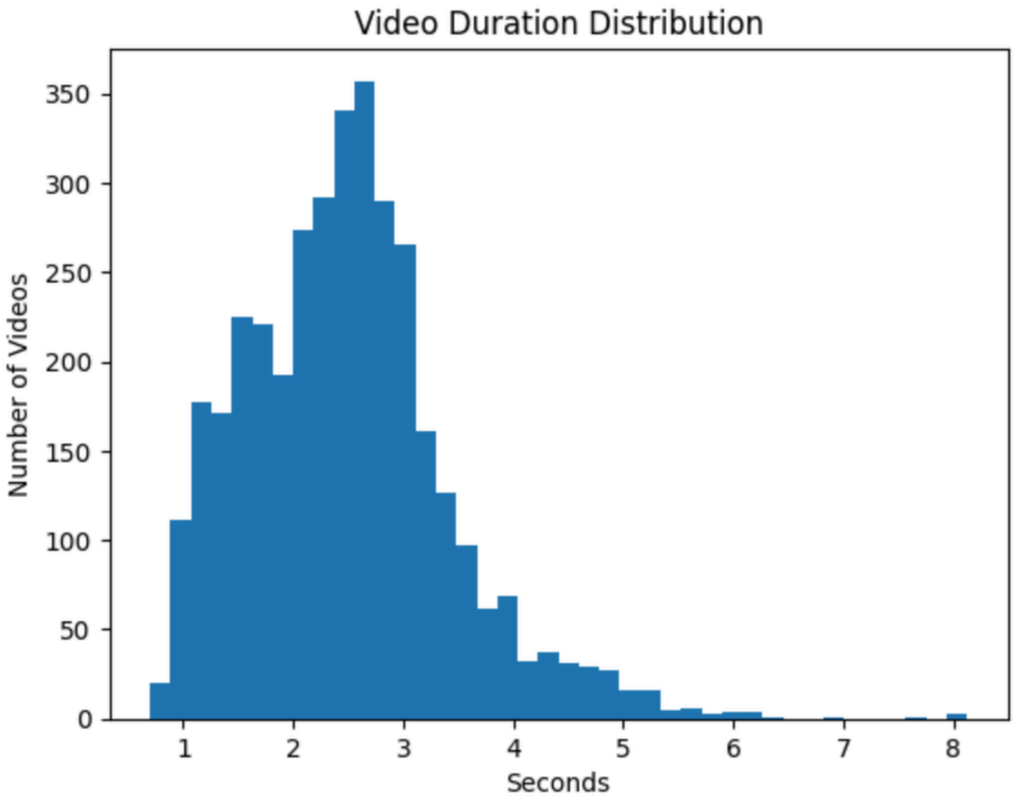
AVERAGE FPS: 28.412864978598535
MIN FPS: 12.0
MAX FPS: 50.0

DATASET STATISTICS

WLASL (100 CLASSES)



WLASL (300 CLASSES)



THANK YOU